

Developing the Pauli Matrices



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Abstract of Presentation

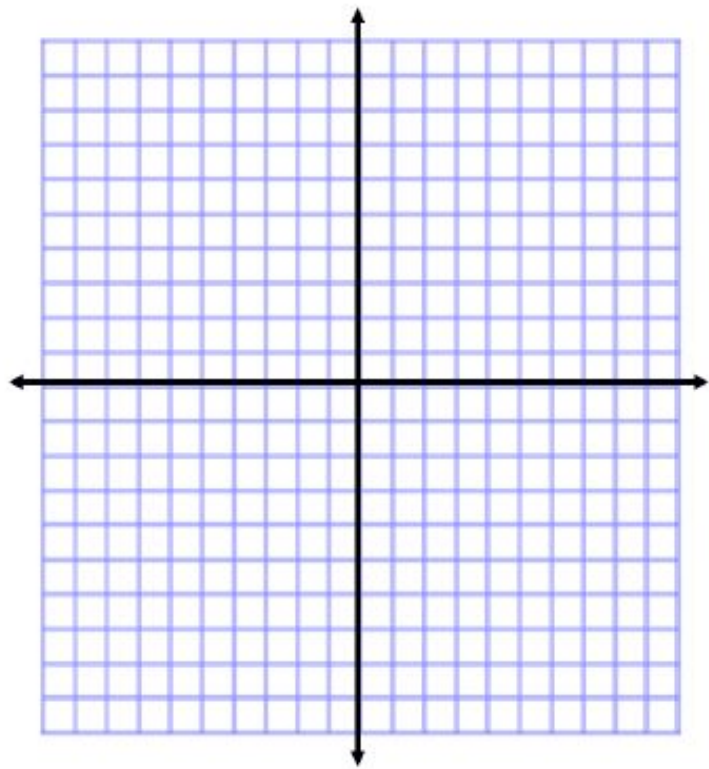
- Develop the Pauli Matrices, which were a famous result in quantum mechanics
 - Reference: Theoretical Minimum Lecture Series by Leonard Susskind
- Understand the basic differences between classical and quantum mechanics
- Highlight the motivation for some of the mathematics in a non-rigorous way

States of Systems

Classical States

$$C = \{H, T\}$$

$$D = \{1, 2, 3, 4, 5, 6\}$$



Classical State: Measurement



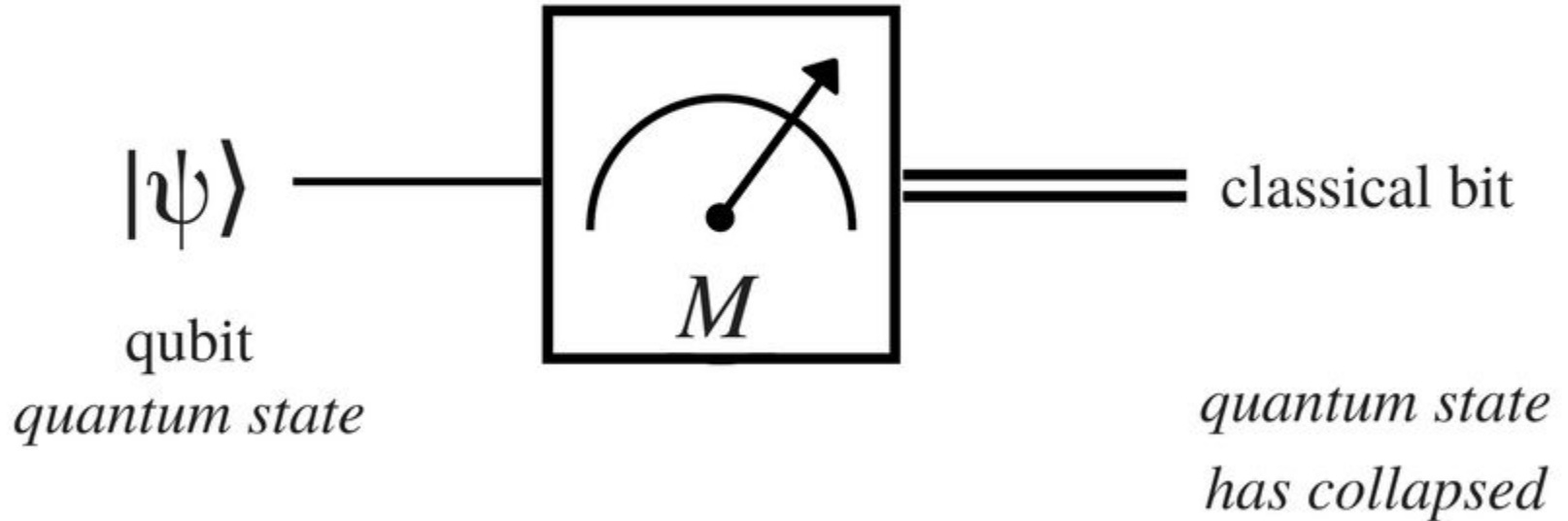
Quantum States

???

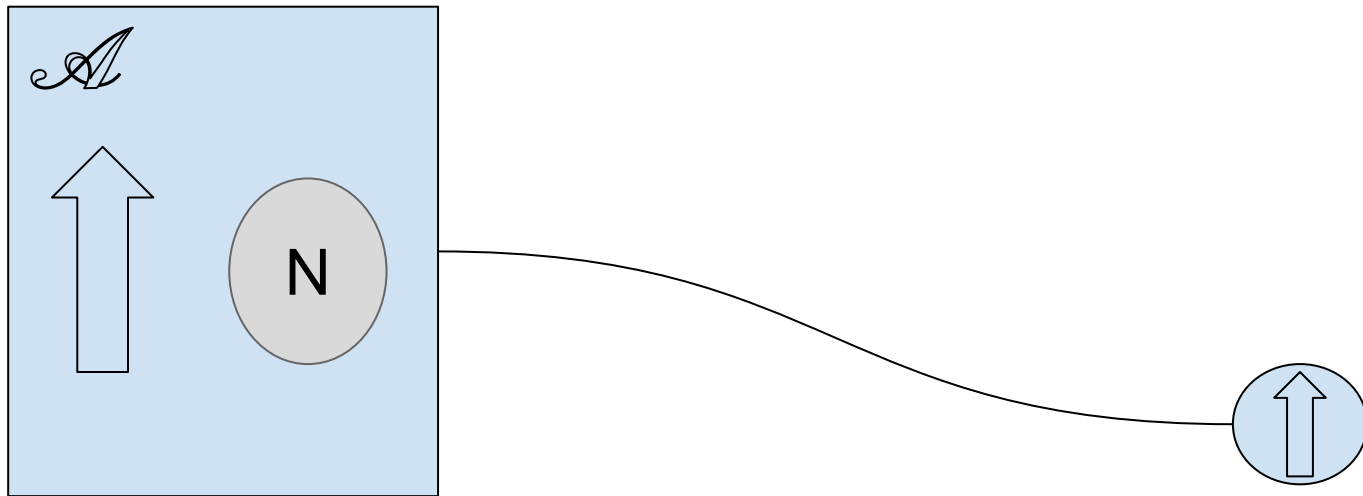
Quantum States: Dirac Notation

$$\langle \Psi | \quad | \Psi \rangle$$

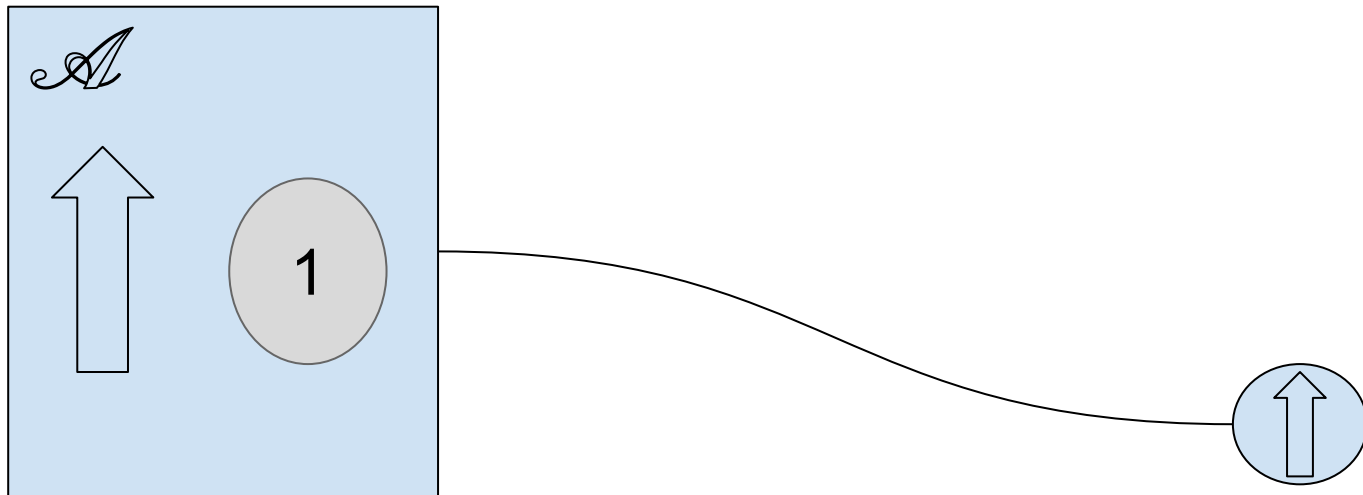
Quantum States: Measurement



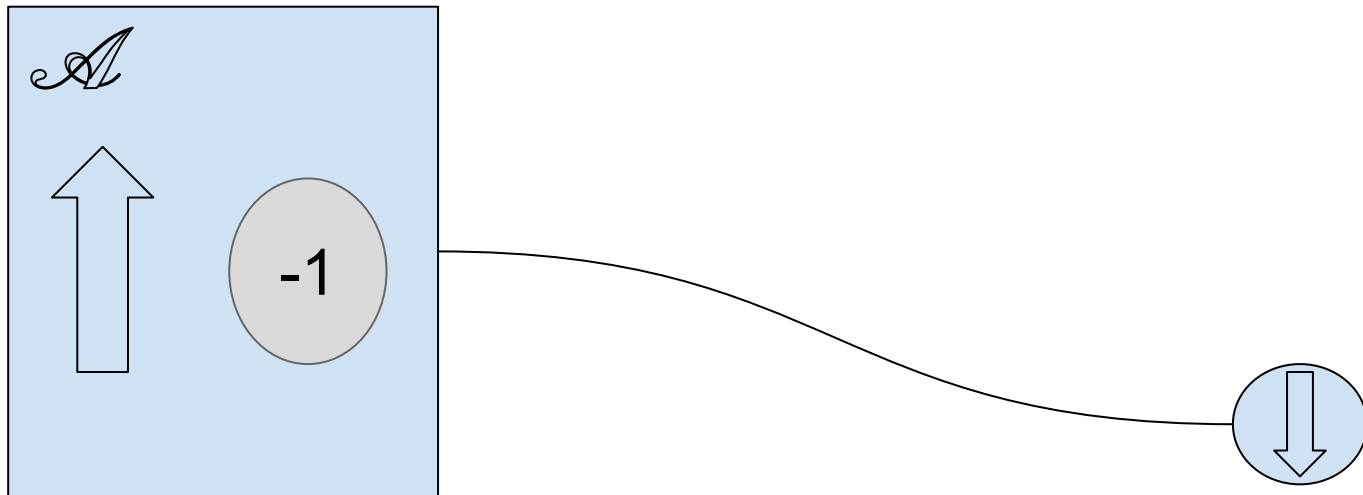
Quantum States: Example



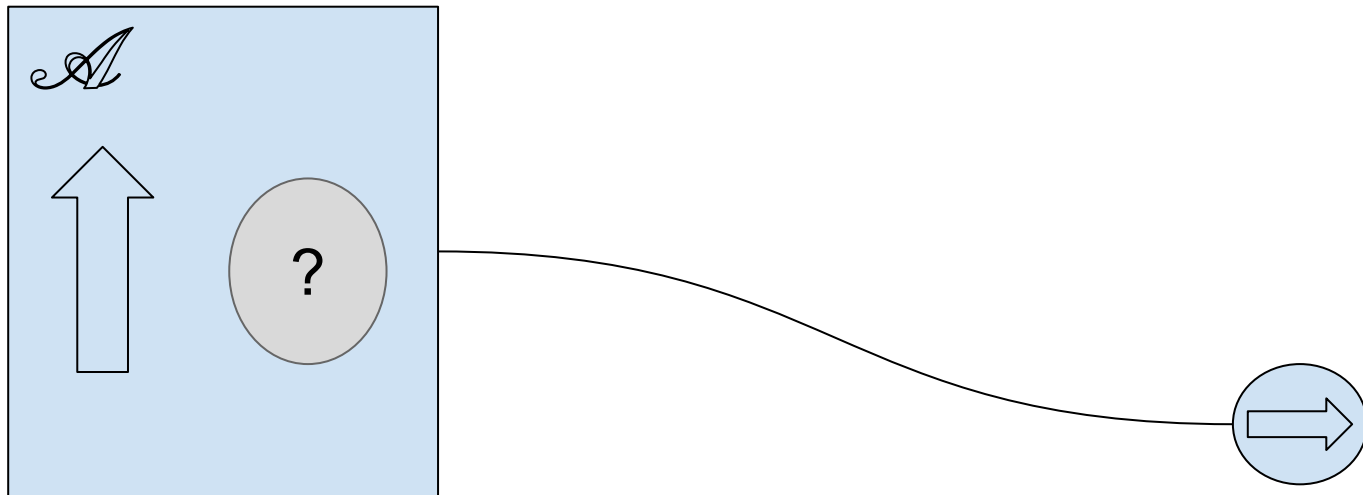
Quantum States: Example



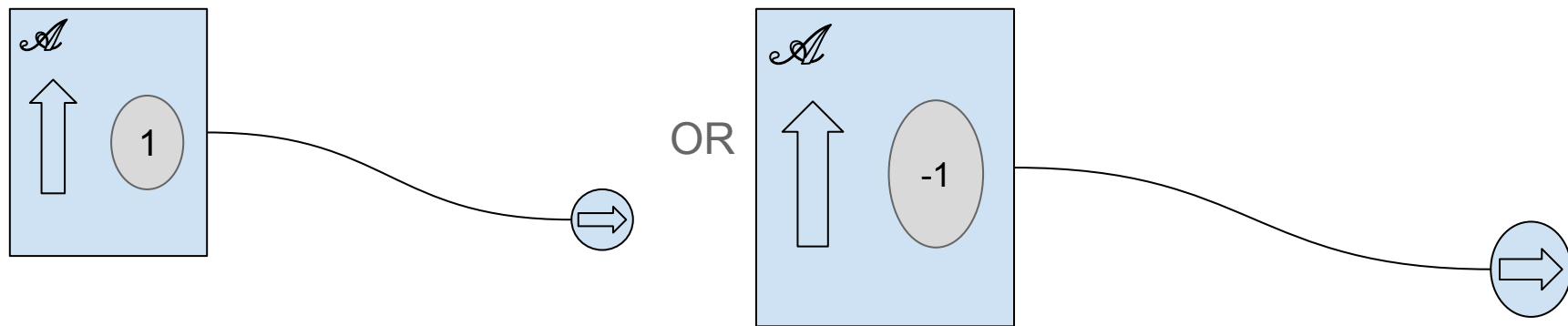
Quantum States: Example



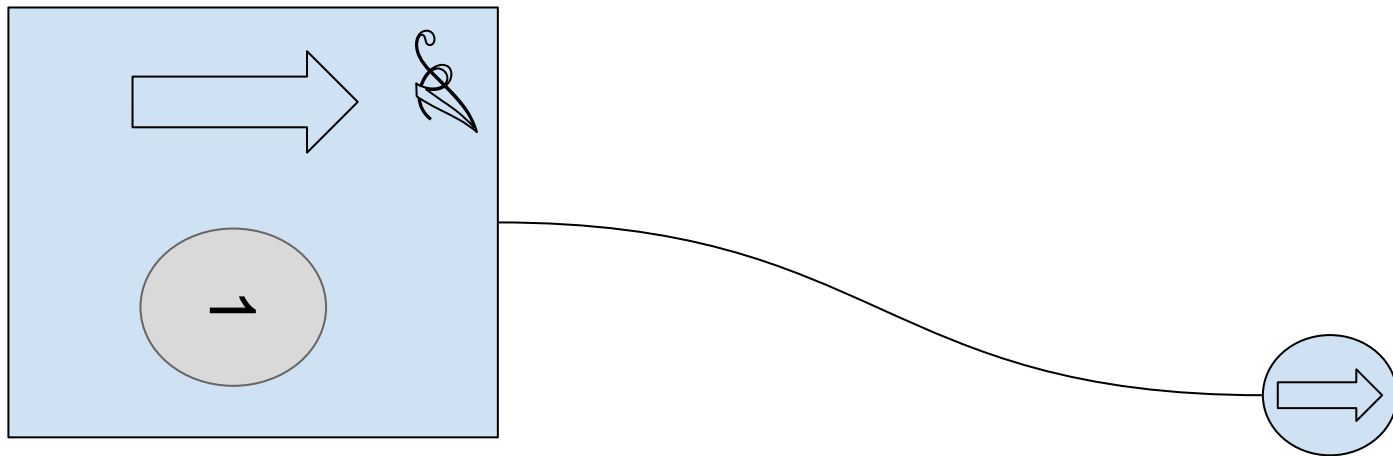
Quantum States: Example



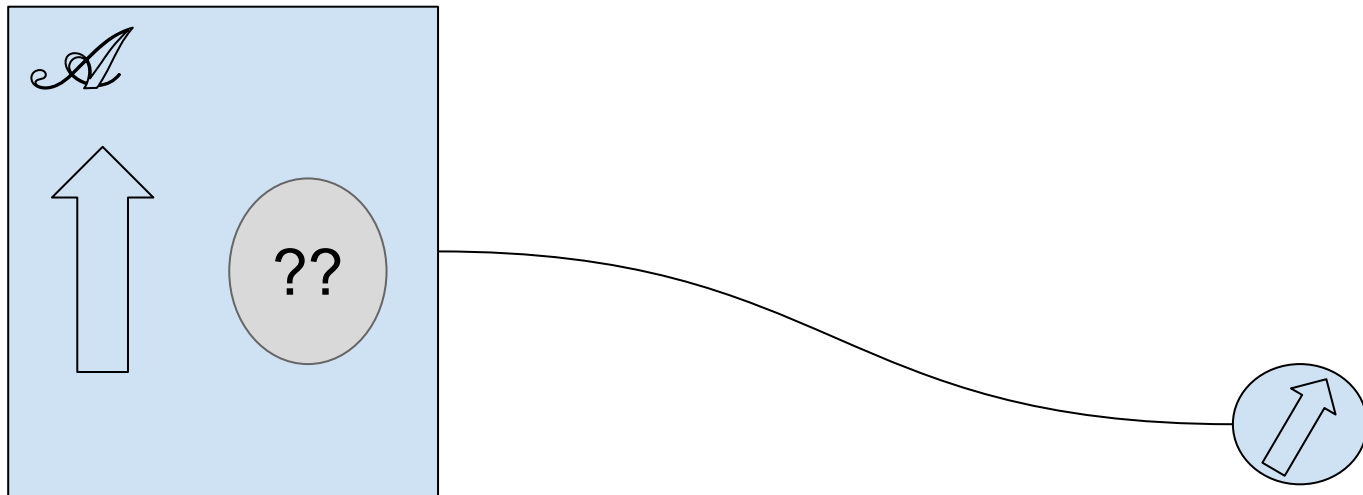
Quantum States: Example



Quantum States: Example



Quantum States: Example



Quantum Mechanics: Operators

Let σ_z denote the apparatus that measures the spin in the up direction

Let $|u\rangle$ denote upward spin

Let $|d\rangle$ denote the downward spin

$$\langle u|d\rangle = \langle d|u\rangle = 0$$

$$\sigma_z|u\rangle = 1 \text{ and } \sigma_z|d\rangle = -1?$$

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Representing a Quantum Object

$$|\Psi\rangle = \alpha_u|u\rangle + \alpha_d|d\rangle$$

The Right Vector

$$|r\rangle = \alpha_u|u\rangle + \alpha_d|d\rangle$$

The Right Vector

$$|r\rangle = \frac{1}{\sqrt{2}} |u\rangle + \frac{1}{\sqrt{2}} |d\rangle$$

The Left Vector

$$|l\rangle = \frac{1}{\sqrt{2}} |u\rangle - \frac{1}{\sqrt{2}} |d\rangle$$

The In and Out Vectors

$$|I\rangle = \frac{1}{\sqrt{2}} |u\rangle + \frac{i}{\sqrt{2}} |d\rangle$$

$$|O\rangle = \frac{1}{\sqrt{2}} |u\rangle - \frac{i}{\sqrt{2}} |d\rangle$$

Verifying probabilities

$$|\langle u|r\rangle|^2 = \left|\frac{1}{\sqrt{2}}\langle u|u\rangle + \frac{1}{\sqrt{2}}\langle u|d\rangle\right|^2 = \frac{1}{2}$$

$$|\langle r|I\rangle|^2 = \left|\frac{1}{\sqrt{2}}\langle r|u\rangle + \frac{i}{\sqrt{2}}\langle r|d\rangle\right|^2 = \frac{1}{2}$$

Calculating σ_z

$$A_{ij} = \hat{e}_i A \hat{e}_j$$

$$(\sigma_z)_{ij} = \langle i | \sigma_z | j \rangle$$

Calculating σ_z

$$\langle u | \sigma_z | u \rangle = \langle u | (1 | u \rangle) = 1(\langle u | u \rangle) = 1$$

$$\langle d | \sigma_z | u \rangle = \langle d | (1 | u \rangle) = 1(\langle d | u \rangle) = 0$$

$$\langle u | \sigma_z | d \rangle = \langle u | (-1 | d \rangle) = -1(\langle u | d \rangle) = 0$$

$$\langle d | \sigma_z | d \rangle = \langle d | (-1 | d \rangle) = -1(\langle d | d \rangle) = -1$$

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Calculating the other Matrices

$$\langle u | \sigma_x | u \rangle = \frac{1}{\sqrt{2}} \langle u | (\sigma_x | r \rangle + \sigma_x | l \rangle) = \frac{1}{\sqrt{2}} (\langle u | r \rangle - \langle u | l \rangle) = 0$$

$$\langle d | \sigma_x | u \rangle = \frac{1}{\sqrt{2}} \langle d | (\sigma_x | r \rangle + \sigma_x | l \rangle) = \frac{1}{\sqrt{2}} (\langle d | r \rangle - \langle d | l \rangle) = 1$$

$$\langle u | \sigma_x | d \rangle = \frac{1}{\sqrt{2}} \langle u | (\sigma_x | r \rangle - \sigma_x | l \rangle) = \frac{1}{\sqrt{2}} (\langle u | r \rangle + \langle u | l \rangle) = 1$$

$$\langle d | \sigma_x | d \rangle = \frac{1}{\sqrt{2}} \langle d | (\sigma_x | r \rangle - \sigma_x | l \rangle) = \frac{1}{\sqrt{2}} (\langle d | r \rangle + \langle d | l \rangle) = 0$$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\langle u | \sigma_y | u \rangle = \frac{1}{\sqrt{2}} (\langle u | (\sigma_y | I \rangle + \sigma_y | O \rangle) = \frac{1}{\sqrt{2}} (\langle u | I \rangle - \langle u | O \rangle) = 0$$

$$\langle d | \sigma_y | u \rangle = \frac{1}{\sqrt{2}} (\langle d | (\sigma_y | I \rangle + \sigma_y | O \rangle) = \frac{1}{\sqrt{2}} (\langle d | I \rangle - \langle d | O \rangle) = i$$

$$\langle u | \sigma_y | d \rangle = \frac{1}{i\sqrt{2}} (\langle u | (\sigma_y | I \rangle - \sigma_y | O \rangle) = \frac{1}{i\sqrt{2}} (\langle u | I \rangle + \langle u | O \rangle) = \frac{1}{i} = -i$$

$$\langle d | \sigma_y | d \rangle = \frac{1}{i\sqrt{2}} (\langle d | (\sigma_y | I \rangle - \sigma_y | O \rangle) = \frac{1}{i\sqrt{2}} (\langle d | I \rangle + \langle d | O \rangle) = 0$$

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

The Pauli Matrices

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Thank you!